MULTIMODAL RAG TEST DATASET

Renewable Energy Systems: A Synthetic Research and Operations Report

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Purpose: Demonstration document for testing document ingestion, OCR, table extraction, chart understanding, multimodal retrieval, source citations, and question answering.

Important data notice

All figures, regional values, project names, operational observations, and examples in this report are synthetic and created for software testing. They are not official statistics, forecasts, investment advice, or claims about real energy projects.

Prepared for AI document-intelligence and Multimodal RAG testing

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How to use this file

Upload this PDF into your Multimodal RAG application. Test questions that require retrieving a value from a table, interpreting a chart, connecting a narrative with a table, comparing regions, and citing exact page numbers. Because the document contains deliberately labeled synthetic data, it is safe to use as a software test fixture.

1. Executive Summary

This report provides a deliberately varied test corpus for an AI document-intelligence application. It combines explanatory prose, tables, chart images, a workflow diagram, project records, maintenance logs, budget assumptions, risk descriptions, and a glossary. The content is designed to exercise retrieval across different representations rather than only plain paragraphs.

The synthetic dataset models renewable energy projects across five fictionalized regions: West, South, North, East, and Central. It includes solar, wind, hydro, and bioenergy categories. The numerical series are internally consistent for demonstration but do not represent real-world measurements.

Testing objectives

• Retrieve specific facts from long documents and cite the source page.

• Read structured tables without mixing values between rows or columns.

• Interpret trends shown in embedded charts.

• Combine information from narrative explanations and numerical tables.

• Recognize uncertainty and avoid claiming that synthetic data is official.

• Return a clear insufficient-evidence response when a question is not answered by the report.

Key dataset characteristics

Attribute	Description
Topic	Renewable energy planning and operations
Data status	Entirely synthetic; created for application testing
Modalities	Text, tables, charts, diagram, page-level references
Time coverage	Illustrative annual series from 2018 through 2025
Regional categories	West, South, North, East, Central
Technology categories	Solar, wind, hydro, bioenergy
Recommended tests	Retrieval, cross-modal reasoning, citations, abstention

2. Scope and Dataset Notes

This document is a test fixture, not a scientific publication. Names of regions and project records are used as categories for structured retrieval. Values were invented for this report to create meaningful comparisons and test cases. No claim should be made that a synthetic value is an actual installed capacity, generation total, cost, emissions reduction, or employment figure.

Units and interpretation

Unless explicitly stated otherwise, values labelled “index” or “synthetic units” are arbitrary demonstration values. The capacity index is not measured in megawatts. Cost figures are illustrative budget units and must not be interpreted as Indian rupees or market quotations. The purpose is to test whether a RAG application preserves labels and units.

Data handling recommendations

A robust ingestion pipeline should preserve the original document, extracted text, page number, table rows, chart captions, and visual evidence references. It should also record whether a value came from text extraction, OCR, a table parser, or a visual model. When an answer is generated, citations should be traceable to retrieved evidence.

Known limitations

This dataset does not model real grid dispatch, actual energy policy, measured emissions, live market prices, or engineering design limits. It does not contain private personal information. Some sections intentionally repeat concepts with different wording to test retrieval ranking and source selection.

3. Technology Overview

Renewable energy systems convert naturally replenished energy flows into useful electricity or heat. This synthetic report uses four categories to exercise document classification: solar, wind, hydro, and bioenergy. A practical planning workflow may consider resource availability, connection capacity, land constraints, equipment reliability, financing, and environmental safeguards.

Solar systems

Solar photovoltaic systems convert incident sunlight into electricity. Output can vary with time of day, season, cloud cover, panel orientation, shading, and equipment condition. In this dataset, the solar capacity index increases over the illustrative period, but the index is only a synthetic trend for retrieval tests.

Wind systems

Wind turbines convert kinetic energy in moving air into electricity. Production depends on wind speed, turbine characteristics, wake effects, availability, and grid curtailment. The synthetic wind series grows more gradually than the solar series.

Hydro systems

Hydropower uses flowing or stored water to drive turbines. Actual output can be affected by rainfall, reservoir levels, ecological requirements, and seasonal water availability. The synthetic hydro series is comparatively stable.

Bioenergy systems

Bioenergy may use organic residues or other biomass resources. Its feasibility depends on feedstock supply, logistics, conversion efficiency, air-emissions controls, and competing uses for the feedstock. The synthetic bioenergy index changes only modestly in the example series.

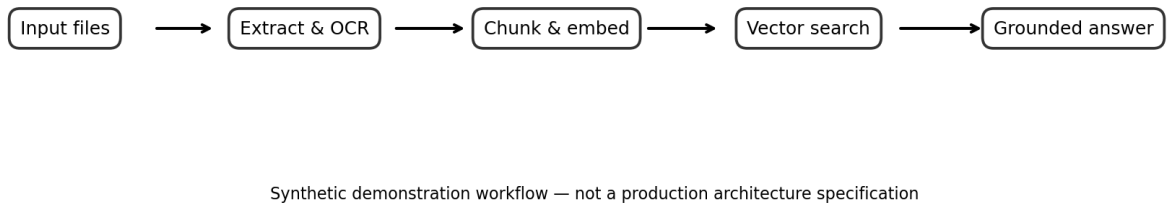


Figure 3.1: Simplified document-intelligence workflow included for multimodal testing.

4. Synthetic Capacity Dataset

Table 4.1 contains the annual technology series used in Figure 4.1. A system should retain the year and technology labels when retrieving values. For example, the solar index in 2022 is 63, while the wind index in the same year is 42. These are synthetic index values, not physical capacity units.

Year	Solar index	Wind index	Hydro index	Bioenergy index
2018	24	35	45	10
2019	35	37	46	10
2020	42	39	46	11
2021	50	40	47	11
2022	63	42	47	12
2023	74	45	48	12
2024	91	48	49	13
2025	108	51	50	13

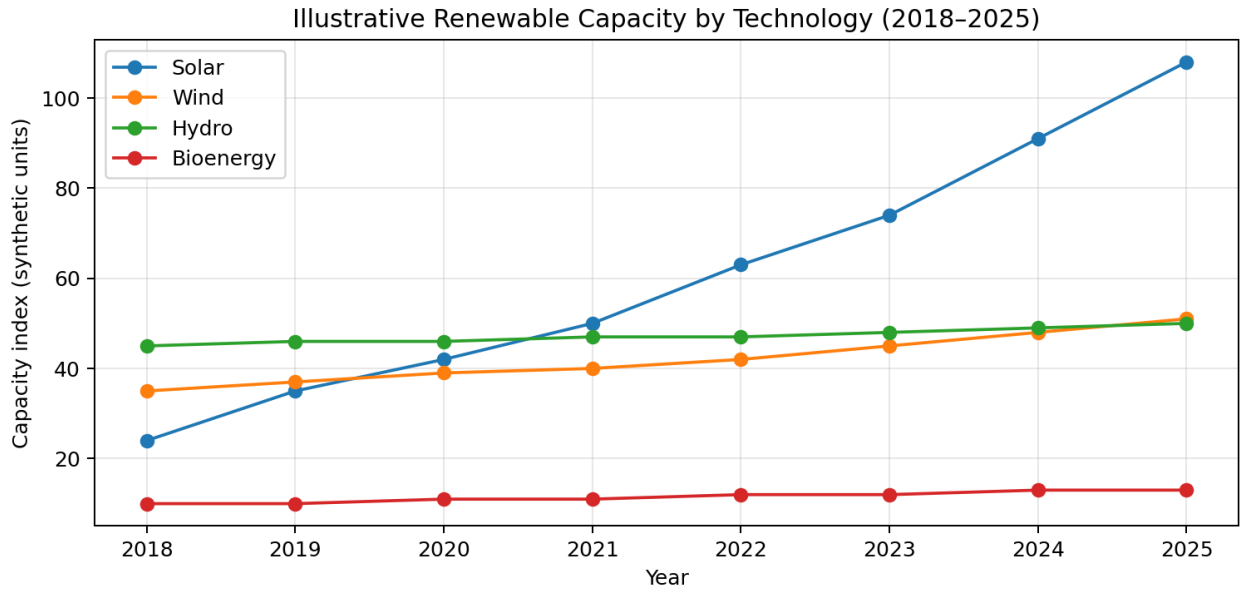


Figure 4.1: Synthetic capacity trends by technology, 2018–2025.

Interpretation for test purposes: solar shows the strongest increase in this series; wind rises more slowly; hydro remains relatively stable; and bioenergy shows a small upward movement. An answer should identify the period and the metric before describing the trend.

5. Regional Comparison

The regional capacity index is a synthetic summary designed for table and chart retrieval. The “West” category has the highest listed capacity index in this dataset, at 92, followed by South at 85. East has the lowest listed index, at 39. These values should not be generalized to real regions.

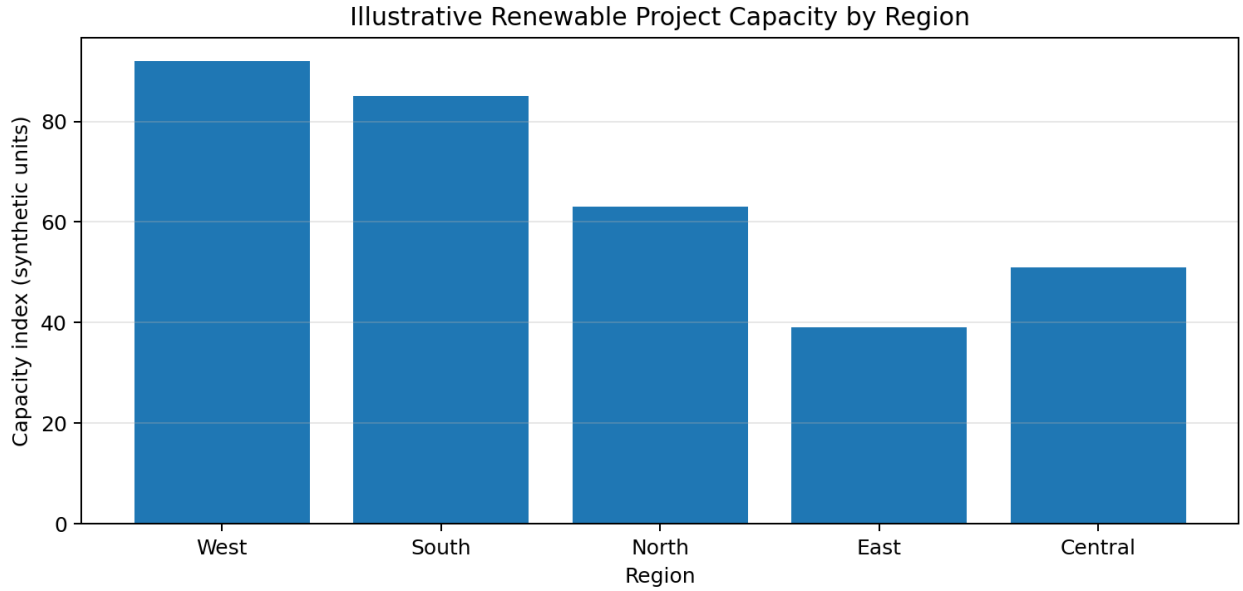


Figure 5.1: Illustrative renewable project capacity index by region.

Region	Capacity index	Estimated jobs index (thousands)	Example focus	Planning note
West	92	18.5	Solar and hybrid	Grid connection and land screening
South	85	16.2	Wind and solar	Seasonality and transmission
North	63	11.8	Solar and hydro	Weather and terrain constraints
East	39	7.1	Bioenergy and solar	Feedstock logistics and distribution
Central	51	9.4	Solar and hybrid	Network capacity and maintenance

Cross-modal test: ask which region has the highest index, then ask whether the chart and table support the same ranking. The answer should cite both sources if the application uses both.

6. Quarterly Generation

Table 6.1 contains quarterly generation index values for solar and wind. The series is intentionally different from the annual capacity series. Capacity and generation are not interchangeable concepts, so a system must not use one table to answer a question about the other.

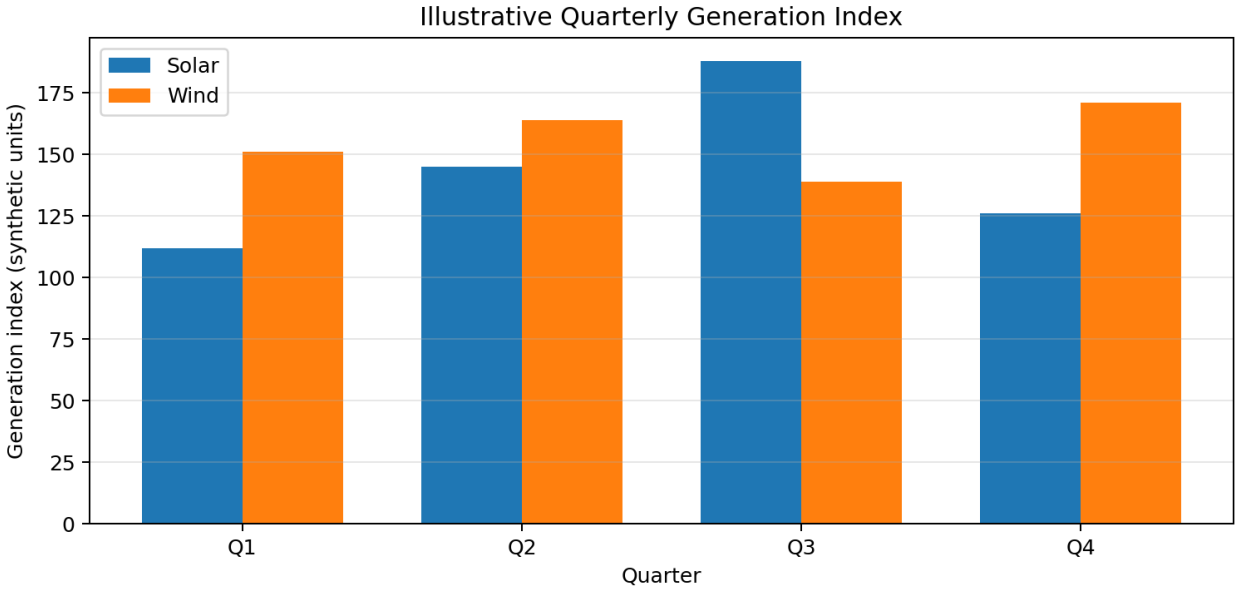


Figure 6.1: Quarterly generation index, synthetic units.

Quarter	Solar generation index	Wind generation index	Higher value
Q1	112	151	Wind
Q2	145	164	Wind
Q3	188	139	Solar
Q4	126	171	Wind

The highest solar quarterly index is 188 in Q3. The highest wind quarterly index is 171 in Q4. Wind is higher than solar in Q1, Q2, and Q4; solar is higher in Q3. These explicit comparisons can be used to test table-based reasoning and chart interpretation.

7. Project Register

The following records represent a fictional project portfolio. The IDs are unique test identifiers. Capacity, cost, schedule, and risk scores are invented values. The register helps test exact lookup, filtering, comparisons, and multi-hop questions.

Project ID	Project name	Region	Type	Capacity index	Budget units	Status	Risk score
RE-001	East Hydro Project 01	East	Hydro	25	35	Design	3
RE-002	South Hybrid Project 02	South	Hybrid	42	58	Under construction	5
RE-003	Central Hydro Project 03	Central	Hydro	59	81	Operational	2
RE-004	East Hybrid Project 04	East	Hybrid	13	17	Under review	4
RE-005	South Wind Project 05	South	Wind	30	40	Design	1
RE-006	Central Hybrid Project 06	Central	Hybrid	47	63	Under construction	3
RE-007	North Wind Project 07	North	Wind	64	86	Operational	5
RE-008	South Bioenergy Project 08	South	Bioenergy	18	22	Under review	2
RE-009	Central Wind Project 09	Central	Wind	35	45	Planning	4
RE-010	North Bioenergy Project 10	North	Bioenergy	52	68	Under construction	1
RE-011	West Solar Project 11	West	Solar	69	91	Operational	3
RE-012	Central Bioenergy Project 12	Central	Bioenergy	23	27	Under review	5
RE-013	North Solar Project 13	North	Solar	40	50	Planning	2
RE-014	West Hydro Project 14	West	Hydro	57	73	Design	4
RE-015	East Solar Project 15	East	Solar	11	96	Operational	1
RE-016	North Hydro Project 16	North	Hydro	28	32	Under review	3
RE-017	West Hybrid Project 17	West	Hybrid	45	55	Planning	5
RE-018	East Hydro Project 18	East	Hydro	62	78	Design	2
RE-019	South Hybrid Project 19	South	Hybrid	16	14	Under construction	4
RE-020	West Wind Project 20	West	Wind	33	37	Under review	1
RE-021	East Hybrid Project 21	East	Hybrid	50	60	Planning	3
RE-022	South Wind Project 22	South	Wind	67	83	Design	5
RE-023	Central Bioenergy Project 23	Central	Bioenergy	21	19	Under construction	2
RE-024	East Wind Project 24	East	Wind	38	42	Operational	4
RE-025	South Bioenergy Project 25	South	Bioenergy	55	65	Planning	1

Suggested questions: What is the status of RE-014? Which listed projects are in the East region? Compare the capacity indices of RE-003 and RE-021. Which projects have a risk score of 5? Answers must be computed from the table rather than guessed.

8. Operations and Maintenance

Operations and maintenance records can reveal repeated failure patterns, response times, and equipment availability. This section contains synthetic event logs. The report does not claim that the listed events occurred in actual facilities.

Event ID	Project ID	Asset	Issue	Severity	Response hours	Resolution	Status
EV-001	RE-008	Battery controller	Sensor calibration	High	4.8	Cleaning completed	Monitoring
EV-002	RE-015	SCADA gateway	Output below baseline	Medium	8.0	Part replaced	In progress
EV-003	RE-022	Pump	Communication loss	Low	11.2	Pending specialist review	Closed
EV-004	RE-004	Panel string	Protective trip	High	14.5	Firmware updated	Closed
EV-005	RE-011	Turbine sensor	Inspection overdue	Medium	17.8	Inspection completed	Monitoring
EV-006	RE-018	Transformer	Unexpected vibration	Low	3.0	Cleaning completed	In progress
EV-007	RE-025	Inverter	Temperature warning	High	6.2	Part replaced	Closed
EV-008	RE-007	Battery controller	Sensor calibration	Medium	9.5	Pending specialist review	Closed
EV-009	RE-014	SCADA gateway	Output below baseline	Low	12.8	Firmware updated	Monitoring
EV-010	RE-021	Pump	Communication loss	High	16.0	Inspection completed	In progress
EV-011	RE-003	Panel string	Protective trip	Medium	19.2	Cleaning completed	Closed
EV-012	RE-010	Turbine sensor	Inspection overdue	Low	4.5	Part replaced	Closed
EV-013	RE-017	Transformer	Unexpected vibration	High	7.8	Pending specialist review	Monitoring
EV-014	RE-024	Inverter	Temperature warning	Medium	11.0	Firmware updated	In progress
EV-015	RE-006	Battery controller	Sensor calibration	Low	14.2	Inspection completed	Closed
EV-016	RE-013	SCADA gateway	Output below baseline	High	17.5	Cleaning completed	Closed
EV-017	RE-020	Pump	Communication loss	Medium	2.8	Part replaced	Monitoring
EV-018	RE-002	Panel string	Protective trip	Low	6.0	Pending specialist review	In progress
EV-019	RE-009	Turbine sensor	Inspection overdue	High	9.2	Firmware updated	Closed
EV-020	RE-016	Transformer	Unexpected vibration	Medium	12.5	Inspection completed	Closed

For a useful test, ask for the highest-severity events, average response time, or the event associated with a specified project. If calculations are supported, the application should state which records were included in the calculation.

9. Cost and Finance Tables

The cost table uses arbitrary budget units. They are not rupees, dollars, or market estimates. Use the numbers only to test numerical extraction, totals, ratios, and comparisons.

Cost category	Scenario A	Scenario B	Scenario C	Notes
Equipment	420	455	438	Panels, turbines, conversion equipment
Civil works	105	122	118	Foundations and site preparation
Grid connection	76	91	84	Illustrative connection costs
Monitoring	18	22	20	Sensors and monitoring systems
Training	9	12	11	Operator training
Maintenance reserve	32	39	35	Illustrative contingency
Environmental safeguards	14	17	16	Monitoring and mitigation
Other	21	24	23	Miscellaneous

Scenario totals can be calculated by summing the categories: Scenario A = 695 units, Scenario B = 782 units, and Scenario C = 745 units. Scenario B has the largest illustrative total. These totals are provided as answer checks for evaluation.

Cost interpretation rules

Do not infer that a lower initial budget automatically means lower lifecycle cost. Actual comparison would require operating costs, financing, project life, degradation, replacement schedules, taxes, and other assumptions. This dataset intentionally lacks those real-world inputs.

10. Environmental Indicators

This section contains invented environmental indicators for testing structured retrieval. Values are normalized demonstration indices and must not be interpreted as actual carbon savings or environmental measurements.

Scenario	Land-use index	Water-use index	Lifecycle impact index	Community consultation score / 10	Monitoring frequency
Solar A	42	12	28	8.1	Quarterly
Wind B	31	9	24	7.6	Monthly
Hydro C	58	45	36	7.2	Monthly
Bioenergy D	37	33	41	6.8	Weekly
Hybrid E	44	18	29	8.4	Quarterly

In this synthetic table, Wind B has the lowest lifecycle impact index at 24. Hybrid E has the highest community consultation score at 8.4 out of 10. Hydro C has the highest water-use index at 45. These answers should preserve the metric names.

Environmental review checklist

• Define the measurement boundary and units.

• Record the method used to estimate each indicator.

• Separate measured values from modelled or illustrative values.

• Document uncertainty and missing observations.

• Include stakeholder consultation and monitoring records.

• Avoid treating a single index as a complete environmental assessment.

11. Risk Register

The risk register is designed to test retrieval across risk descriptions, severity, likelihood, mitigation, and ownership. Scores are synthetic ordinal values and are not the output of a validated risk model.

Risk ID	Risk	Likelihood (1–5)	Impact (1–5)	Owner role	Mitigation	Trigger
RK-01	Grid connection delay	4	4	Project manager	Confirm milestones with network operator	Milestone missed
RK-02	Equipment supply delay	3	4	Procurement lead	Qualify alternate suppliers	Lead time rises
RK-03	Inverter overheating	3	5	Operations lead	Thermal inspection and alarms	Temperature threshold
RK-04	Poor wind availability	2	3	Resource analyst	Review resource assumptions	Output below baseline
RK-05	Feedstock shortage	4	4	Supply coordinator	Diversify supply contracts	Inventory below reserve
RK-06	OCR extraction error	3	2	AI platform owner	Review low-confidence pages	Low OCR confidence
RK-07	Incorrect source citation	2	5	AI platform owner	Validate citations against retrieval IDs	Missing source ID
RK-08	Unauthorized document access	2	5	Security lead	Enforce document-level access checks	Access violation
RK-09	Unexpected maintenance	3	3	Maintenance lead	Track recurring events	Repeated alert
RK-10	Incorrect table interpretation	3	4	Data engineer	Validate headers and row boundaries	Conflicting extracted values

A multimodal RAG assistant should distinguish a listed risk from an observed event. For example, RK-08 describes a possible security risk; it does not assert that an unauthorized access incident has occurred.

12. Policy and Implementation Scenarios

The following fictional scenarios are narrative examples designed to test questions that connect prose with structured evidence.

Scenario A: Capacity expansion

A planning team is comparing the regional capacity index with a portfolio of project proposals. The team notes that a high index does not by itself prove that a region is suitable for every new project. Transmission capacity, site constraints, environmental safeguards, and community consultation still require separate evaluation.

Scenario B: Reliability improvement

An operations team observes repeated communication loss and output-below-baseline alerts. It proposes to combine event logs with equipment histories and maintenance records. The objective is to find recurring patterns and prioritize inspections, not to allow an AI model to make unreviewed safety-critical decisions.

Scenario C: Document intelligence

A researcher uploads a technical report with charts, scanned pages, and cost tables. A text-only search may retrieve a paragraph while missing the graph itself. A multimodal retrieval system should retain the relation between the narrative and visual evidence, then cite the relevant page and evidence type.

Scenario D: Data governance

An organization wants staff to ask questions over private documents. The application must enforce document access boundaries, protect secrets, support deletion, and avoid sending unnecessary document content to external providers. Model-generated answers must not override the organization's access rules.

Scenario E: Evaluation before deployment

A development team builds a small evaluation set with known answers and supporting pages. It measures retrieval relevance, citation validity, latency, and the ability to abstain when the evidence is missing. A polished interface alone is not evidence that the retrieval system works.

13. Case Studies

Case Study 1: Finding a value in a chart

Question: In the annual capacity chart, which technology has the highest index in 2025? Expected answer: Solar, with an index of 108. Supporting evidence: Table 4.1 and Figure 4.1. A complete answer should state that the value is synthetic.

Case Study 2: Comparing seasonal generation

Question: In which quarter does solar exceed wind, and what are the values? Expected answer: Q3; solar is 188 and wind is 139. Supporting evidence: Table 6.1 and Figure 6.1.

Case Study 3: Joining risk and system architecture

Question: Which listed risk addresses incorrect table interpretation, and what mitigation is recorded? Expected answer: RK-10; validate headers and row boundaries. The question tests exact table retrieval and preservation of the mitigation field.

Case Study 4: Distinguishing cost from capacity

Question: Which budget scenario has the highest total, and does that mean it has the highest capacity? Expected answer: Scenario B has the largest illustrative total at 782 units. The report does not provide a direct mapping from these budget scenarios to project capacity, so the second part cannot be concluded from the supplied data.

Case Study 5: Recognizing unsupported claims

Question: What was India's actual renewable electricity production in 2025 according to this report? Expected answer: The report does not provide actual national production figures; its data is synthetic and not official statistics.

14. Cross-Modal Questions and Findings

This section lists reasoning patterns that a multimodal RAG application should support. Answers should cite source evidence and preserve the distinction between synthetic test data and real-world facts.

Test type	Question	Expected answer / evaluation target
Trend comparison	Which technology shows the strongest increase from 2018 to 2025?	Solar; its synthetic index rises from 24 to 108.
Regional ranking	Which region has the highest capacity index?	West, with an index of 92.
Quarterly comparison	When is solar's quarterly generation index higher than wind's?	Q3 only, with 188 versus 139.
Exact lookup	What is the recorded mitigation for RK-07?	Validate citations against retrieval IDs.
Source limitation	Are the report's figures official government data?	No. The report explicitly labels all values as synthetic.
Evidence insufficiency	What is the real-world cost of building a 100 MW solar farm in Pune?	The report does not contain this real-world cost information.
Cross-section reasoning	Why should a high regional capacity index not be treated as proof that every project is suitable?	The scenario says transmission capacity, site constraints, safeguards, and consultation require separate evaluation.
Data distinction	Are capacity and generation interchangeable in this report?	No. The report explicitly warns that they are different concepts.
Visual citation	What is the 2025 solar index shown in the annual chart?	108, supported by Table 4.1 and Figure 4.1.
Risk classification	Does RK-08 prove that a security incident occurred?	No. It describes a possible risk, not an observed incident.

Evaluation recommendation: score answer correctness separately from citation correctness. An answer can contain a correct value but still fail if it cites the wrong page or invents evidence.

15. Glossary

Term	Definition
Capacity index	A synthetic indicator used in this report to compare illustrative capacity values.
Generation index	A synthetic indicator used to compare illustrative energy-generation values across quarters.
OCR	Optical Character Recognition; a method for extracting text from images or scanned pages.
Embedding	A numerical representation used to compare semantic similarity between items.
Vector database	A system that stores embeddings and supports similarity search.
Chunk	A bounded portion of content used for indexing and retrieval.
Multimodal retrieval	Retrieval that can use more than one content modality, such as text and images.
Context fusion	Combining retrieved evidence into a coherent context for answer generation.
Grounded answer	An answer tied to identifiable evidence rather than unsupported model output.
Citation validation	Checking that a source reference actually exists and supports the associated claim.
Hallucination	Generated content that is unsupported or incorrect despite being presented confidently.
Abstention	A response that clearly states when available evidence is insufficient.
Cross-modal reasoning	Reasoning that combines evidence from different modalities.
Latency	The elapsed time required to complete an operation.
Synthetic dataset	Data created for testing or demonstration rather than collected as real-world observations.

16. Appendix: Full Synthetic Records

This appendix provides additional records to increase the size and diversity of the searchable corpus. Values are synthetic. The records can be used for filtering, counting, ranking, and exact lookup tests.

Record ID	Region	Technology	Year	Capacity index	Availability %	Maintenance events	Notes
AR-001	North	Bioenergy	2019	29	82.7	3	Weather-related variance
AR-002	Central	Hydro	2020	48	83.4	6	Monitoring update
AR-003	North	Solar	2021	67	84.1	0	Equipment review
AR-004	Central	Hybrid	2022	86	84.8	3	Grid coordination
AR-005	South	Hydro	2023	14	85.5	6	Routine inspection
AR-006	Central	Wind	2024	33	86.2	0	Weather-related variance
AR-007	South	Hybrid	2025	52	86.9	3	Monitoring update
AR-008	East	Bioenergy	2018	71	87.6	6	Equipment review
AR-009	South	Wind	2019	90	88.3	0	Grid coordination
AR-010	East	Solar	2020	18	89.0	3	Routine inspection
AR-011	West	Bioenergy	2021	37	89.7	6	Weather-related variance
AR-012	East	Hydro	2022	56	90.4	0	Monitoring update
AR-013	West	Solar	2023	75	91.1	3	Equipment review
AR-014	North	Hybrid	2024	94	91.8	6	Grid coordination
AR-015	West	Hydro	2025	22	92.5	0	Routine inspection
AR-016	North	Wind	2018	41	93.2	3	Weather-related variance
AR-017	Central	Hybrid	2019	60	93.9	6	Monitoring update
AR-018	North	Bioenergy	2020	79	94.6	0	Equipment review
AR-019	Central	Wind	2021	98	95.3	3	Grid coordination
AR-020	South	Solar	2022	26	96.0	6	Routine inspection
AR-021	Central	Bioenergy	2023	45	96.7	0	Weather-related variance
AR-022	South	Hydro	2024	64	97.4	3	Monitoring update
AR-023	East	Solar	2025	83	98.1	6	Equipment review
AR-024	South	Hybrid	2018	11	98.8	0	Grid coordination
AR-025	East	Hydro	2019	30	82.5	3	Routine inspection
AR-026	West	Wind	2020	49	83.2	6	Weather-related variance
AR-027	East	Hybrid	2021	68	83.9	0	Monitoring update
AR-028	West	Bioenergy	2022	87	84.6	3	Equipment review
AR-029	North	Wind	2023	15	85.3	6	Grid coordination
AR-030	West	Solar	2024	34	86.0	0	Routine inspection
AR-031	North	Bioenergy	2025	53	86.7	3	Weather-related variance
AR-032	Central	Hydro	2018	72	87.4	6	Monitoring update
AR-033	North	Solar	2019	91	88.1	0	Equipment review
AR-034	Central	Hybrid	2020	19	88.8	3	Grid coordination
AR-035	South	Hydro	2021	38	89.5	6	Routine inspection
AR-036	Central	Wind	2022	57	90.2	0	Weather-related variance
AR-037	South	Hybrid	2023	76	90.9	3	Monitoring update

Record ID	Region	Technology	Year	Capacity index	Availability %	Maintenance events	Notes
AR-038	East	Bioenergy	2024	95	91.6	6	Equipment review
AR-039	South	Wind	2025	23	92.3	0	Grid coordination
AR-040	East	Solar	2018	42	93.0	3	Routine inspection

Important: availability percentages in this appendix are synthetic and may be used to test sorting or filtering only. Do not treat them as operational records from actual power plants.

17. Data Quality and Provenance

A document-intelligence system should preserve provenance at every transformation stage. The original file is the primary artifact. Extracted text is a derived representation, OCR output is a derived representation, and model-generated captions are also derived content. Each representation should retain a reference to the original page or image. This enables users to inspect evidence when extraction quality is uncertain.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

18. Retrieval Design Considerations

A retrieval system can combine lexical search and semantic similarity. Lexical search can be useful for exact identifiers such as RE-014 or RK-07, while semantic retrieval can be useful for questions phrased differently from the source text. A practical system may use metadata filters, hybrid retrieval, reranking, and deduplication. The retrieval design should be evaluated using known questions and supporting passages.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

19. Handling Tables and Charts

Tables encode relationships through rows, columns, headers, and units. Extracting a table as an unordered stream of words can cause a model to associate a number with the wrong label. Charts encode relationships through axes, legends, colors, bars, and lines. A robust pipeline should preserve chart images and captions and use a suitable visual model when answering questions that depend on visual structure.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

20. Uncertainty and Safe Answering

When evidence is missing, conflicting, or low quality, the system should say so. It should not fill a gap with a plausible but unsupported number. When a scanned page has low OCR confidence, the system can flag the source for review. When multiple sources disagree, the answer should describe the disagreement and cite each source instead of silently choosing one.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

21. Performance and Resource Management

Document processing may require CPU time, memory, model API calls, and vector storage. Systems should avoid repeating expensive operations unnecessarily. Ingestion can be asynchronous, and processing status should be visible. Caching may be useful when safe, but caches must respect document access boundaries and deletion requests.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

22. Security and Access Boundaries

Uploaded documents may contain confidential information. A production application should authenticate users, authorize each document operation, validate files, limit resource consumption, and keep API secrets server-side. Prompt injection inside a document should be treated as untrusted content rather than a command that can override the application's system instructions.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

23. Suggested Demonstration Procedure

Upload this report, wait until indexing finishes, and ask an exact lookup question such as the 2025 solar index. Then ask a chart interpretation question, a question joining the risk register with the narrative, and an unsupported real-world question. Verify both the answer and the cited source. Finally, delete the document and check that its searchable content is no longer available.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

24. Conclusion

This synthetic report is designed to exercise the full path from ingestion to retrieval and answer generation. It contains varied representations and deliberately explicit limitations. A successful implementation should retrieve the correct evidence, preserve units and labels, answer questions requiring more than one source, cite the correct page, and acknowledge when the document cannot answer a question.

Implementation check

Confirm that the application records the source page, extracted content type, processing status, and retrieval metadata. For questions involving visual content, verify that the visual evidence is actually passed to a capable model or that the system clearly states the limitation.

25. Evaluation Questions for RAG Testing

Use the following questions after uploading the PDF. The answer key can be used to evaluate the application, but the application should independently retrieve the evidence before answering.

No.	Question	Expected answer
1	What is the 2025 solar capacity index?	108
2	What is the 2025 wind capacity index?	51
3	Which region has the lowest capacity index?	East, 39
4	What is the highest solar quarterly generation index and which quarter?	188 in Q3
5	What is the highest wind quarterly generation index and which quarter?	171 in Q4
6	Which risk concerns incorrect table interpretation?	RK-10
7	What mitigation is listed for RK-08?	Enforce document-level access checks
8	Which cost scenario has the largest total?	Scenario B, 782 synthetic units
9	Are the cost units actual currency?	No, they are arbitrary budget units
10	Does the report provide actual national renewable electricity production for 2025?	No
11	Which technology series is most stable in the annual table?	Hydro, in the provided series
12	Which region has a capacity index of 51?	Central
13	What is the mitigation for RK-07?	Validate citations against retrieval IDs
14	Does RK-08 mean an unauthorized access incident occurred?	No
15	What should the system do when evidence is insufficient?	State that the available evidence is insufficient
16	Which scenario has the highest community consultation score?	Hybrid E, 8.4/10
17	Which scenario has the highest water-use index?	Hydro C, 45
18	When does solar generation exceed wind generation?	Q3
19	What is the 2025 bioenergy index?	13
20	Can the report determine the real-world cost of a 100 MW solar farm in Pune?	No

End of synthetic test report. All figures and records in this document are invented for testing and must not be cited as real-world statistics.